

# WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY FOR SELF-IMPROVING AGENTS IN ADAPTIVE WORLDS

**Anonymous authors**

Paper under double-blind review

## ABSTRACT

Self-improving agents choose replacement operations with cheap verifiers, yet deployment can change transitions, rewards, and other agents’ behavior. We define *Improvement Fidelity* for an update  $\tau = (\pi_t, \pi_{t+1})$ : proxy improvement  $\Delta_V$  should preserve the sign and ordering of deployment improvement  $\Delta_*$ . PIVOT (Paired Interventional Validation of Optimization Transitions) estimates the differential correction  $\Delta_* - \Delta_V$  with paired high-fidelity (HF) interventions and allocates a fixed query budget by a one-step knowledge-gradient rule per unit cost.

Across controlled and sealed response worlds, we separate three phenomena that are often conflated: deployment response, update-level fidelity error, and decision-changing error. Large response can sharply increase proxy–deployment discrepancy without creating an allocation advantage. In 90 sealed Leduc, Kuhn, and Melting Pot roots, proxy and deployment top-1 updates differ in 51 roots; nevertheless, the descriptive normalized PIVOT–Uniform gain is not larger on flipped roots. The primary allocation contrast is positive in Leduc (0.036, 95% CI [0.014, 0.064]), near-null in Kuhn, and unresolved in Melting Pot, with mixed secondary conditions. A sealed-query audit finds that PIVOT’s query set intersects the proxy/deployment top-1 pair in most roots, but the queried noise relative to the proxy top-2 margin can remain large. Thus response magnitude is not decision relevance, and decision relevance is not validation value: PIVOT-KG can help when HF evidence resolves uncertainty that matters for the update decision.

## 1 INTRODUCTION

Self-improvement is usually written as a loop: generate candidates, score them, promote one, and repeat. The loop consumes a *replacement operation*, not an isolated policy value. In a trading system, a larger order changes liquidity; in an auction, a bid changes rivals’ responses; in a recommender, a policy changes future user behavior. A verifier can therefore rank policies correctly while ranking improvements incorrectly. High-fidelity deployment tests can expose the error, but they are often too costly to run on every candidate.

We study the transition  $\tau_t = (\pi_t, \pi_{t+1})$ . Let  $V$  be a cheap observer or replay world and let  $*$  denote the policy-induced deployment world. Define

$$\Delta_V = J_V(\pi') - J_V(\pi), \quad \Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi). \quad (1)$$

An improvement reversal occurs when  $\Delta_V > 0$  but  $\Delta_* < 0$ . More generally, a fidelity error matters to selection only when it changes a candidate ordering or displaces the no-update action. Figure 1 shows the basic sign failure.

We distinguish three response layers. The *observer/direct* layer holds the incumbent world fixed; the *actor* layer lets the environment react mechanically to the candidate; the *strategic* layer also lets other agents adapt. These are increasingly demanding interventions, not claims that one simulator is universal ground truth.

An improvement operator  $\mathcal{A}$  induces a transition law  $Q_{\mathcal{A}}$  with density  $q_{\mathcal{A}}$ . For loss  $L$ , our estimand is operator-relative Improvement Fidelity,

$$\text{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi, \pi') \sim Q_{\mathcal{A}}} [L(\Delta_V, \Delta_*)]. \quad (2)$$

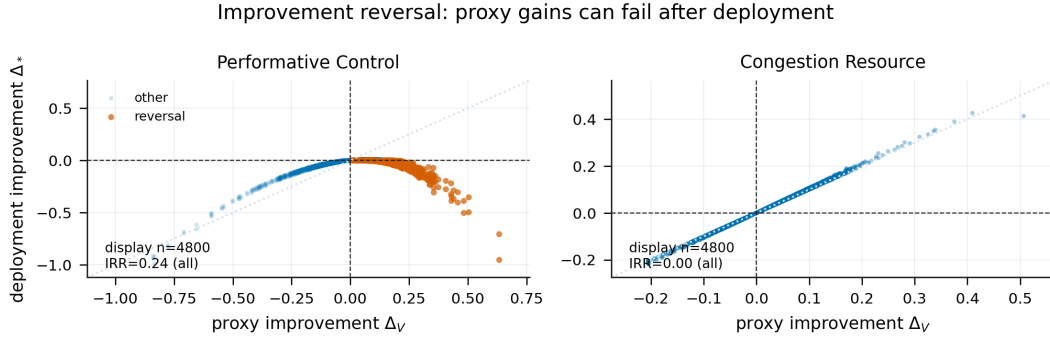


Figure 1: Paired proxy and deployment improvements in the two controlled worlds. The lower-right quadrant is an improvement reversal. Points are a deterministic display subsample; the source bundle retains all transition rows and reports the full-row IRR.

The relevant population is therefore the updates an agent actually proposes. Our contributions are:

**Contribution 1.** We formalize Improvement Fidelity and give six results linking global accuracy, operator shift, response footprint, decision preservation, paired variance, and finite-sample selection.

**Contribution 2.** We introduce PIVOT-KG, a paired differential validator whose one-step knowledge-gradient acquisition maximizes expected reduction in update-selection regret per HF cost under a fixed budget.

**Contribution 3.** We connect theory to controlled and sealed evidence through the chain response  $\rightarrow$  fidelity error  $\rightarrow$  decision consequence  $\rightarrow$  allocation value, retaining positive, null, and negative cohorts rather than asserting universal superiority.

## 2 RELATED WORK

**Performative and strategic response.** Performative prediction and multi-agent extensions make deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2022; 2023); Basu et al. (2025); Piliouras & Yu (2022). They study stable policies, equilibria, or convergence; we ask whether a particular replacement remains an improvement after its response is realized.

**World models and policy evaluation.** WorldGym, policy-aware simulator learning, and interactive market simulators evaluate values, rankings, or trajectory realism Quevedo et al. (2025); Dann et al. (2026); Hansen et al. (2024); Noble et al. (2026); Zhang et al. (2026). Global value fidelity can coexist with a wrong local update sign, which is the gap measured here.

**Self-improvement and verification.** Executable policy evolution and agent benchmarks make iterative evaluation operational Mao et al. (2026); Wang et al. (2026); Zhang et al. (2025); Wei et al. (2026); Guo et al. (2026); Wu et al. (2024). Our verifier is fixed and external: failure can arise because it evaluates the wrong response world, even when the test itself is correct.

**Adaptive selection and paired evaluation.** Active learning, knowledge gradients, and best-arm identification allocate expensive observations Settles (2009); Lindley (1956); Frazier et al. (2009); Chick & Inoue (2001); Kalyanakrishnan et al. (2012). PIVOT changes the loss from global prediction error to update-selection regret and uses paired rollouts to cancel common-mode variation. OpenSpiel, Melting Pot, and finance testbeds provide response mechanisms with different claim boundaries Lanctot et al. (2019); Leibo et al. (2021); Byrd et al. (2019); Liu et al. (2022).

## 3 PROBLEM FORMULATION

**Policy-induced worlds.** Write the deployment response as a response map  $R_{-i}(\pi)$  for other agents, and write the induced deployment world as  $\mathcal{M}[\pi]$ , including any environment response induced by  $\pi$ , with value

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \quad (3)$$

We use three diagnostic layers:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \quad (4)$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \quad (5)$$

$$\Delta_{\text{strategic}} = J_i(\pi', R_{-i}(\pi')) - J_i(\pi, R_{-i}(\pi)). \quad (6)$$

Here  $R_{-i}$  is a general response map; a best-response operator is a special case, while adaptive populations and learning opponents are also covered. The layers are estimands implemented by separate worlds, not an assertion that a learned or observational proxy is ground truth.

**Metrics and scale.** For transition rows,  $\text{IDE} = n^{-1} \sum_k |\Delta_{V,k} - \Delta_{*,k}|$ , and ISC is sign agreement after removing ties. The improvement reversal rate is  $\text{IRR} = P(\Delta_* < 0 \mid \Delta_V > 0)$ ; the strategic analogue is  $\text{SIRR} = P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$ . For a candidate set  $C_t$ ,

$$\text{ISR}_t = \max_{j \in C_t} \Delta_{*,t,j} - \Delta_{*,t,\hat{j}}, \quad (7)$$

and cumulative true improvement is  $\text{CTI}_T = \sum_t \Delta_{*,t,\hat{j}}$ . We define cumulative improvement-selection regret as  $\text{CISR}_T = \sum_t \text{ISR}_t$ ; the budget experiment has  $T = 1$ , so its CISR equals ISR for one candidate set. ISR, CTI, and CISR are in native environment reward units, and we never compare their raw magnitudes across environments. When Proxy Only and all-high-fidelity are matched within the same environment, candidate count, and decision horizon, we may report the fraction of excess regret removed,

$$\text{FER} = 1 - \frac{\text{CISR}_m - \text{CISR}_{\text{allHF}}}{\text{CISR}_{\text{proxy}} - \text{CISR}_{\text{allHF}}}; \quad (8)$$

otherwise FER is withheld.

**Value Fidelity versus Improvement Fidelity.** Policy-level value error and update-level error are different estimands. We use “false improvement (improvement reversal)” only for the event  $\Delta_V > 0$  and  $\Delta_* < 0$ .

## 4 THEORY

The following results are elementary but delimit what a fidelity claim can mean. Proofs are included to make the assumptions auditable.

**Proposition 1** (Global fidelity is sufficient). *If  $\sup_{\pi} |J_V(\pi) - J_*(\pi)| \leq \varepsilon$ , then  $|\Delta_V - \Delta_*| \leq 2\varepsilon$ . The sign is preserved whenever  $|\Delta_*| > 2\varepsilon$ .*

*Proof.*  $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$ ; the triangle inequality gives the bound and the sign statement.  $\square$

**Proposition 2** (Global fidelity blindness). *For every  $\varepsilon > 0$ , there are a finite policy family, a verifier, a policy distribution, and an update operator with policy MAE and rank error below  $\varepsilon$ , but  $\text{IRR} = 1$  and  $\text{ISC} = 0$  on the operator’s transition law.*

*Proof.* Take  $n$  policies with  $J_*(i) = i/(n-1)$  and swap one adjacent pair in  $V$ . Then  $\text{MAE} = 2/[n(n-1)]$  and  $1 - \text{Spearman} = 12/[n(n^2-1)]$ , both tending to zero. Let the operator sample only the swapped directed transition and propose the lower-valued true policy. Its proxy sign is positive and its deployment sign is negative.  $\square$

**Proposition 3** (Operator Shift Bound (centered shift bound)). *Let  $P$  be a calibration law, let  $Q_A \ll P$  have density ratio  $w = q_A/p$ , and let  $\ell(\tau) = L(\Delta_V, \Delta_*)$  have finite variance under  $P$ . Then*

$$\mathbb{E}_{Q_A}[\ell] \leq \mathbb{E}_P[\ell] + \sqrt{\text{Var}_P(\ell) \chi^2(Q_A \| P)}. \quad (9)$$

*Proof.* Write  $\mathbb{E}_Q[\ell] - \mathbb{E}_P[\ell] = \mathbb{E}_P[(w-1)(\ell - \mathbb{E}_P[\ell])]$ . Cauchy–Schwarz and  $\mathbb{E}_P[(w-1)^2] = \chi^2(Q \| P)$  give the result.  $\square$

**Proposition 4** (Response–Footprint Bound). *Suppose  $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_M d(\pi, \pi')$  and the value functional is  $L_J$ -Lipschitz in its world argument. Then*

$$|\Delta_{\text{actor}} - \Delta_{\text{direct}}| \leq L_J L_M d(\pi, \pi'). \quad (10)$$

*Proof.* The incumbent term cancels because both comparisons use  $\mathcal{M}[\pi]$  as the reference. The candidate term is bounded by  $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi])$ .  $\square$

**Proposition 5** (Decision Preservation Under Differential Error). *For candidates  $i, j$  with true margin  $m = \Delta_{*,i} - \Delta_{*,j} > 0$ , if each estimate has absolute error below  $m/2$ , then the estimated ordering is unchanged.*

*Proof.* Subtract the two estimates. The perturbation of the margin has absolute value below  $m$ , so the estimated margin remains positive.  $\square$

**Proposition 6** (Finite-Sample Best-Update Identification). *For every competitor  $j$ , suppose the paired sample advantage of the true best candidate over  $j$  is sub-Gaussian with variance proxy  $2\sigma^2$ . If the best of  $K$  candidates has margin at least  $m > 0$ , then  $n$  paired rollouts per candidate give wrong-selection probability at most  $(K-1) \exp[-nm^2/(4\sigma^2)]$ . It suffices that  $n \geq 4\sigma^2 m^{-2} \log((K-1)/\delta)$  for error at most  $\delta$ .*

*Proof.* For each competitor, a wrong ordering requires the difference of two sample mean errors to exceed  $m$ . Apply the sub-Gaussian tail bound and union bound over  $K-1$  competitors.  $\square$

**Lemma 1** (Paired variance). *For a paired rollout difference  $Y' - Y$ ,*

$$\text{Var}(Y' - Y) = \text{Var}(Y') + \text{Var}(Y) - 2 \text{Cov}(Y', Y). \quad (11)$$

*Thus positive common-mode covariance lowers the variance of the improvement estimate. The finite-sample proposition treats each paired best-versus- $j$  advantage as sub-Gaussian with variance proxy  $2\sigma^2$ ; it is a finite-sample assumption, not a claim about every deployment world.*

The first two propositions show that global value fidelity is sufficient but not necessary. The operator-shift bound says that local fidelity can deteriorate when the update operator concentrates away from calibration. The response bound motivates measuring an update footprint, while decision preservation identifies the relevant scale: an error matters when it is large relative to a candidate margin. The paired-variance lemma explains why common randomness makes a finite HF budget more informative. Strategic responses need not be globally Lipschitz; that is an empirical diagnostic.

## 5 PIVOT-KG

The theory identifies the target of validation: queries should resolve uncertainty at the candidate margin, while high-fidelity evaluation is too expensive to apply indiscriminately. Given an incumbent and  $K$  candidates, PIVOT-KG therefore computes cheap proxy deltas and an update footprint  $z_\Delta$ . The footprint can include policy distance, occupancy shift, action/support shift, entropy change, or, for a finance diagnostic, participation and liquidity consumption.

For each candidate  $c$ , write the differential correction as  $G_c = \Delta_{*,c} - \Delta_{V,c}$ . The frozen candidate-level implementation places a joint Gaussian prior over the correction vector,

$$G \sim \mathcal{N}(\mu, K_{\text{prior}}), \quad \Delta_* \mid \Delta_V \sim \mathcal{N}(\Delta_V + \mu, K_{\text{prior}}). \quad (12)$$

The cold-start posterior is fit on a disjoint calibration cohort containing paired HF audit blocks and selection observations. It stores the empirical correction mean, a shrinkage covariance for the candidate correction vector (including mean-estimation uncertainty), and heteroscedastic observation variances estimated from independent paired blocks; the initial candidate mean is therefore  $\Delta_V + \mu$ . For a queried transition, incumbent and candidate share the initial state, exogenous path, random seed, and opponent initialization. Each HF query is incorporated by the same exact Gaussian conditioning update used for acquisition fantasies, while unqueried candidates retain their posterior means. The calibration/test split is fixed before decisions, so no audit label from the held-out decision root enters acquisition.

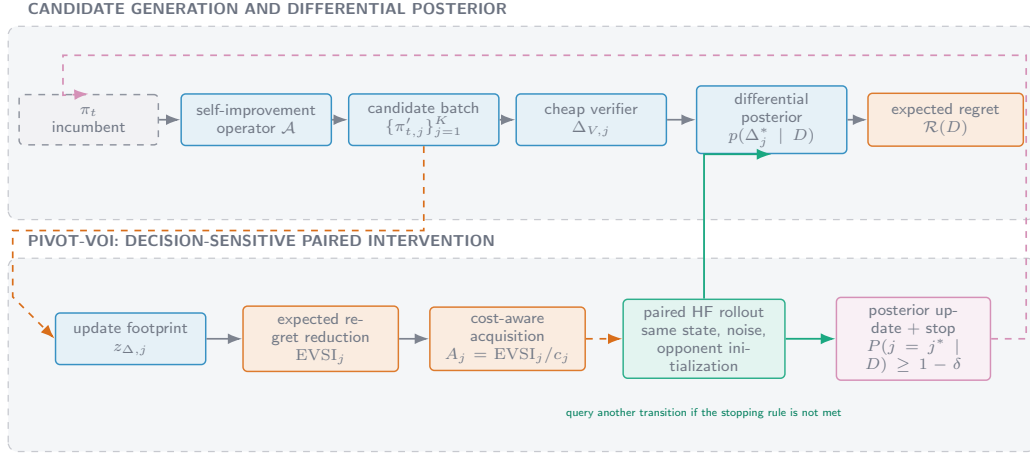


Figure 2: PIVOT-KG’s decision-critical path. Candidate-aware EVSI-per-cost selects paired HF interventions under common state, noise, and opponent initialization; posterior regret determines querying and promotion.

Pairing removes common-mode noise before aggregation. Include the incumbent as an explicit no-update action:  $\mathcal{A} = \{0, 1, \dots, K\}$  with  $\Delta_{*,0} = 0$  and candidates  $j \geq 1$ . Let  $D$  be current paired evidence and  $a(D) = \arg \max_{j \in \mathcal{A}} \mathbb{E}[\Delta_j^* | D]$ . With posterior expected simple regret  $\mathcal{R}(D)$ , a hypothetical HF observation  $Y_j$  has

$$\text{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j|D} \mathcal{R}(D \cup \{Y_j\}), \quad A_j = \text{EVSI}_j / c_j. \quad (13)$$

PIVOT-KG is the one-step knowledge-gradient rule that queries the largest  $A_j$  under a fixed budget in the primary comparison. Optional stopping is a secondary variant. Queried values replace predictions; unqueried candidates retain their proxy/model estimates. The target is update-selection regret.

**Why Transition Validation Differs from Active Learning.** Active learning usually chooses a label that reduces global prediction loss. PIVOT-KG chooses a paired query that can change the sign or ordering of a replacement operation; its loss is update-selection regret.

## 6 EXPERIMENTS

### 6.1 EXPERIMENTAL PROTOCOL

The experiments answer four questions rather than treating each benchmark as a separate win report. Controlled response worlds use paired common-random-number rollouts and 30 independent seeds. The seed, trajectory, opponent-seed cluster, or held-out root is the inferential unit; transition rows are retained for audit and are not treated as independent replicates. The supplied sealed archive contains 30 held-out roots in each of Leduc, Kuhn, and Melting Pot, with 1,800 sealed decisions per cohort. Manuscript generation verifies hashes, seals, and scored-row preservation; it does not rerun an episode.

### 6.2 WHAT BREAKS IMPROVEMENT FIDELITY?

The observer/direct, actor, and strategic layers isolate progressively stronger response mechanisms. The operator-shift study is a complementary failure source: its high-minus-low shift contrast in operator-relative IDE is 0.2627 with bootstrap interval  $[0.1684, 0.3763]$  across 30 seeds and 36000 paired rows (Figure 3). Local update error rises while global policy rank remains comparatively stable. Together these results distinguish distributional operator shift from policy-induced deployment response; either can separate proxy and deployment improvement without implying that every operator or every update fails.

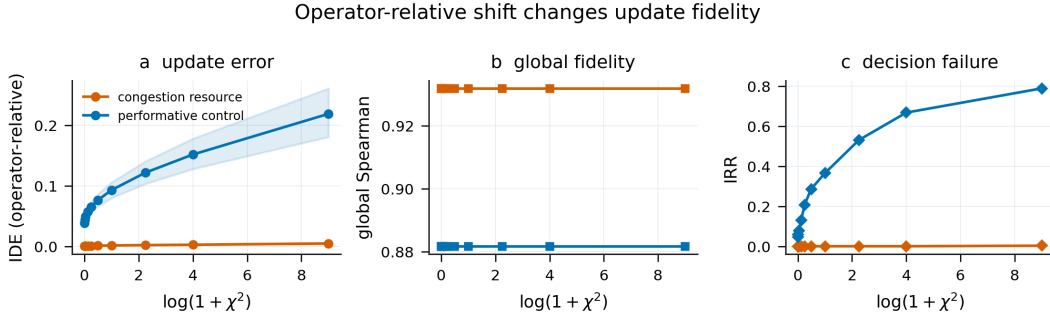


Figure 3: Operator-relative shift diagnostic. Local improvement error rises with registered shift while global policy rank remains comparatively stable. The same environment-by-shift cells show error, rank, and reversal rate.

Table 1: Sealed response-world allocation contrasts. Entries are paired PIVOT-KG minus Uniform gains with root-bootstrap intervals. Primary null and negative conditions are retained as evidence.

| World / condition              | $n$ | PIVOT-Uniform | 95% CI           | PIVOT ISR | Status             |
|--------------------------------|-----|---------------|------------------|-----------|--------------------|
| Leduc, long / 2 queries        | 30  | 0.036         | [0.014, 0.064]   | 0.009     | primary positive   |
| Kuhn, long / 2 queries         | 30  | 0.002         | [0.000, 0.006]   | 0.056     | near-null          |
| Melting Pot, long / 2 queries  | 30  | 0.906         | [-0.784, 3.401]  | 0.836     | primary null       |
| Melting Pot, long / 1 query    | 30  | 3.423         | [0.390, 6.944]   | 1.859     | secondary positive |
| Melting Pot, short / 5 queries | 30  | -0.575        | [-1.042, -0.180] | 1.610     | negative           |

### 6.3 WHEN DOES FIDELITY ERROR CHANGE THE DECISION?

Proposition 5 gives the bridge: if differential error is below half the true candidate margin, the ordering is preserved. For each sealed root we therefore compute the deployment top-1 margin, the proxy-to-deployment correction, and the largest pairwise correction-to-margin ratio. A ratio at least one marks a proxy/deployment top-1 flip; it is a decision-relevance diagnostic, not an allocation outcome.

Across the 90 sealed roots, 51 have a top-1 flip (18 Leduc, 14 Kuhn, and 19 Melting Pot). To avoid comparing reward scales across worlds, Figure 4 reports each root’s squared response gap after division by the square of its largest candidate range across proxy and long/short deployment deltas. The resulting within-root normalized gaps are 0.57 [0.45, 0.70] for Leduc, 0.50 [0.38, 0.62] for Kuhn, and 0.13 [0.10, 0.17] for Melting Pot; native-unit gaps remain in the audit. Yet flip status alone does not predict validation value: normalized allocation gain is 0.075 [0.004, 0.157] on flipped roots and 0.140 [0.049, 0.250] on stable roots. Figure 4 therefore supports the sharper claim: response can create decision-relevant error, but decision relevance is not sufficient for an allocation advantage.

We then ask whether the sealed HF queries could resolve the fragile decision. For each root, the ex-post decision-critical pair is the union of the proxy and deployment top-1 candidates, while the proxy top-2 margin measures how much uncertainty separates the leading proxy candidates. At the primary long budget, the PIVOT query set intersects this pair in 0.97 [0.92, 1.00] of roots (1.00 Leduc, 1.00 Kuhn, and 0.90 Melting Pot). The descriptive normalized gain is 0.113 [0.055, 0.180] among 87 hit roots and -0.195 [-0.584, 0.000] among the three misses. The mean queried HF noise divided by the proxy top-2 margin is 13.99 [7.97, 22.90], and the final sealed estimate separation is 0.06 [0.05, 0.08] after root scaling. These are observable resolvability diagnostics, not posterior selection probabilities or an EVSI estimate: sequential posterior states are not part of the sealed decision rows.

### 6.4 CAN TARGETED HF VALIDATION REDUCE SELECTION REGRET?

Table 1 reports the sealed fixed-budget PIVOT-KG versus Uniform comparison. Leduc is the only cohort with a primary interval strictly above zero: 0.036 [0.014, 0.064]. Kuhn is near-null: 0.002 [0.000, 0.006]. Melting Pot is unresolved in the primary long/two-query condition (0.906 [-0.784, 3.401]); its one-query secondary condition is positive (3.423 [0.390, 6.944]) and its short/five-query condition is negative (-0.575 [-1.042, -0.180]). Its much larger response gap therefore does not imply a positive allocation contrast.

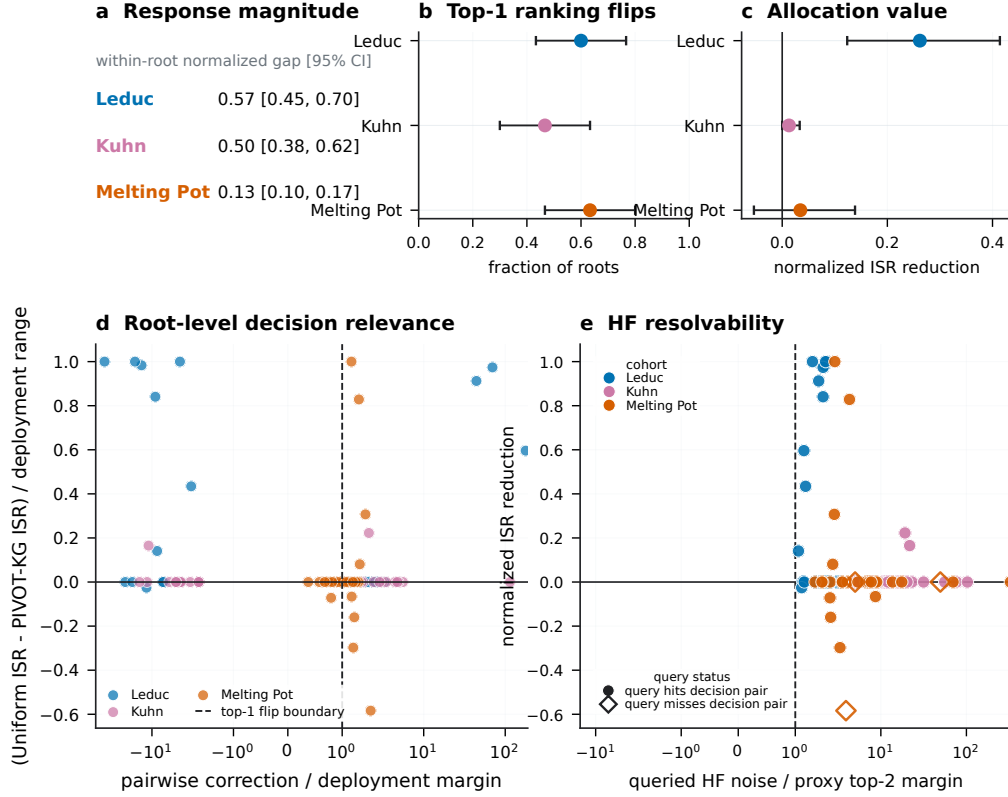


Figure 4: Response magnitude is not allocation value. a reports within-root normalized response gaps; b reports proxy/deployment top-1 flips; c reports normalized ISR reduction; d relates correction-to-margin ratio to normalized gain; e relates queried HF noise to the proxy top-2 margin, with filled circles for queries intersecting the ex-post decision-critical pair and open diamonds for queries that miss it.

The closed-loop study is consistent with this scoped interpretation. Across 40 rounds, PIVOT-KG versus Proxy Only reduces paired CISR by 1.4883 [1.0618, 1.9105], while contrasts against Global-VOI and Paired LUCB are not uniformly favorable. The evidence-efficiency frontier therefore measures a decision rule under matched costs, not universal method ordering.

## 6.5 BOUNDARY RESULTS AND NON-DOMINANCE

The powered OOD transition-evaluator comparison is a negative result rather than a superiority result: transition-minus-global ISC is  $-0.2590$   $[-0.3588, -0.1707]$ , with the interval entirely below zero. The strategic fixture adds an independent adaptive layer and yields  $-0.0240$   $[-0.0249, -0.0231]$ , with family-balanced SIRR 0.9495. These results show why an update-level estimand does not require a particular transition learner to dominate.

## 6.6 STRESS TESTS BEYOND CONTROLLED ENVIRONMENTS

The public finance audit uses historical paths, virtual fills, and a participation/depth proxy. It has 0/7 depth-proxy reversals in the expanded primary sessions and 0/5 in the holdout; the pooled depth-minus-replay effect is approximately  $-4.16 \times 10^{-7}$  in native session units. This is a boundary diagnostic, not causal market impact. The strategic mechanisms are finite fixtures rather than general equilibrium, and external adaptive worlds are endpoint evidence rather than connected trajectories.

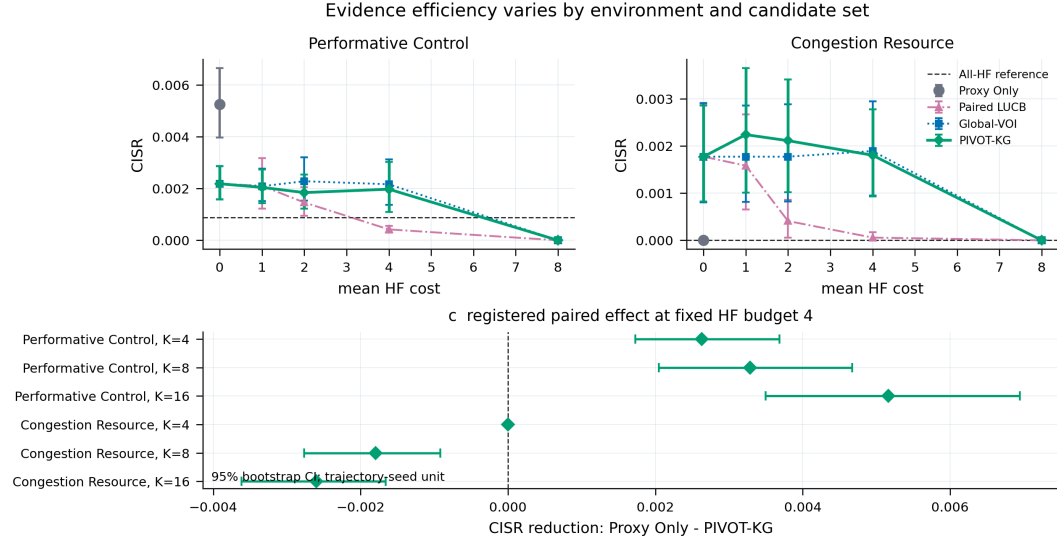


Figure 5: Evidence-efficiency frontier. Matched-cost curves and the fixed-budget forest report selection-regret reduction; the all-HF line is a reference. Intervals are seed bootstraps and no uniform ordering is asserted.

## 7 DISCUSSION AND SCOPE

The evidence supports a conditional claim. Deployment response can increase proxy–deployment error; an error changes the update only when it is large relative to the candidate margin; and PIVOT-KG can help when limited HF queries resolve that decision-relevant uncertainty. The sealed-query audit makes the last condition testable at the level of observable query criticality and residual separation, while stopping short of an unrecorded posterior or EVSI claim. A large response gap, a top-1 flip, or a positive result in one cohort is not by itself an allocation guarantee. The study does not claim universal superiority, simulator ground truth, market causality, or equilibrium behavior.

## 8 CONCLUSION

An agent can be correct about the wrong world. The relevant statistical object for self-improvement is therefore the transition  $\pi \rightarrow \pi'$ , and the core question is whether its sign and ordering survive deployment-induced response. Across controlled and sealed worlds, deployment response separates proxy and deployment improvements, but the size of that separation does not by itself predict a ranking flip or an allocation gain. In the sealed study, 51 of 90 roots changed their proxy/deployment top-1, yet flipped roots did not have a larger normalized PIVOT–Uniform gain.

PIVOT-KG targets the next link in this chain: whether limited HF evidence can resolve a decision-relevant uncertainty before an update is promoted. Its positive Leduc contrast, near-null Kuhn result, and mixed or unresolved Melting Pot conditions support a conditional claim rather than universal superiority. The no-update action and paired evidence are part of the decision problem, so the method can retain the incumbent when all estimated improvements are weak or negative. The claims remain bounded by fixed response worlds, held-out sealed roots, and finite HF budgets; strategic fixtures are not equilibrium results, and the finance audit is not causal market evidence.

The broader implication is that evaluation of self-improvement should validate consequences under the response induced by the proposed update. The resulting evidence ladder is

$$\text{observer} \longrightarrow \text{actor} \longrightarrow \text{strategic}, \quad (14)$$

with each arrow an additional intervention, not an assumed truth claim. Valid progress in adaptive worlds means validating consequences, not merely certifying actions; the value of an HF budget depends on whether that validation can change the update decision.

**AI Use Statement.** Generative AI tools were used for language editing, code and figure-formatting assistance, limited assistance in developing experimental infrastructure, and literature discovery. All experimental designs, mathematical claims, generated code, numerical results, cited sources, and manuscript claims were reviewed and verified by the authors, who take full responsibility for the work.

**Reproducibility Statement.** The anonymous supplement contains deterministic configurations, hash-indexed source evidence, figure tables, and rebuild commands. The public finance audit uses virtual fills and contains no credentials, private data, raw vendor archives, or live orders.

**Ethics Statement.** The study uses controlled virtual environments and public market data only; it involves no human subjects, personal data, live trading, or external authorization.

## REFERENCES

- Debabrota Basu, Udvass Das, Brahim Driss, and Uddalak Mukherjee. Performative policy gradient: Optimality in performative reinforcement learning, 2025. URL <https://arxiv.org/abs/2512.20576>.
- David Byrd, Maria Hybinette, and Tucker Hybinette Balch. Abides: Towards high-fidelity market simulation for ai research. *arXiv preprint arXiv:1904.12066*, 2019. URL <https://arxiv.org/abs/1904.12066>.
- Stephen E. Chick and Koichiro Inoue. New two-stage and sequential procedures for selecting the best simulated system. *Operations Research*, 49(5):732–743, 2001.
- Christoph Dann, Yishay Mansour, and Mehryar Mohri. Theoretical foundations and effective algorithms for policy-aware simulator learning, 2026. URL <https://arxiv.org/abs/2605.29032>.
- Peter Frazier, Warren Powell, and Savas Dayanik. The knowledge-gradient policy for correlated normal beliefs. *INFORMS Journal on Computing*, 21(4):599–613, 2009. doi: 10.1287/ijoc.1080.0314.
- Diandian Guo, Cong Cao, Fangfang Yuan, Yingqi Wang, Yueshan Wang, and Dakui Wang. Self-authored verification is unreliable in heuristic self-improving agents, 2026. URL <https://arxiv.org/abs/2607.24300>.
- Nicklas Hansen, Hao Su, and Xiaolong Wang. Td-mpc2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.16828>.
- Shivaram Kalyanakrishnan, Ambuj Tewari, Peter Auer, and Peter Stone. Pac subset selection in stochastic multi-armed bandits. In *Proceedings of the 29th International Conference on Machine Learning*, pp. 655–662, 2012.
- Marc Lanctot, Edward Lockhart, Jean-Baptiste Lespiau, Dustin Morrill, Finbarr Timbers, Edward Foley, Laurent Sifre, Filippo Albertini, Tommy Steiner, Yoram Naddaf, et al. Openspiel: A framework for reinforcement learning in games, 2019. URL <https://arxiv.org/abs/1908.09453>.
- Joel Z. Leibo, Edgar A. Dueñez-Guzman, Alexander Vezhnevets, John Agapiou, Peter Sunehag, Raphael Koster, Jay Matyas, et al. Scalable evaluation of multi-agent reinforcement learning with melting pot. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 6187–6196. PMLR, 2021. URL <https://proceedings.mlr.press/v139/leibo21a.html>.
- Dennis V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956. doi: 10.1214/aoms/1177728069.
- Xiao-Yang Liu, Ziyi Xia, Jingyang Rui, Jiechao Gao, Hongyang Yang, Ming Zhu, Christina Dan Wang, Zhaoran Wang, and Jian Guo. Finrl-meta: Market environments and benchmarks for data-driven financial reinforcement learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 35, 2022. URL <https://arxiv.org/abs/2211.03107>.
- Debmalya Mandal, Stelios Triantafyllou, and Goran Radanovic. Performative reinforcement learning. *arXiv preprint arXiv:2207.00046*, 2022. URL <https://arxiv.org/abs/2207.00046>.

- Debmalya Mandal, Stelios Triantafyllou, and Goran Radanovic. Performative reinforcement learning. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 23642–23680. PMLR, 2023. URL <https://proceedings.mlr.press/v202/mandal23a.html>.
- Jie Mao, Changlun Li, Xiang Li, Qiqi Duan, Jinhui Yuan, Xiang Liu, Yuyu Luo, Jing Tang, Xiaowen Chu, and Nan Tang. Evoquant: Self-evolving verifier-guided strategy optimization for robust quantitative trading, 2026. URL <https://arxiv.org/abs/2607.12455>.
- Patrick Noble, Mathieu Rosenbaum, and Saad Souilmi. Bridging the reality gap in limit order book simulation. *arXiv preprint arXiv:2603.24137*, 2026. URL <https://arxiv.org/abs/2603.24137>.
- Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dunner, and Moritz Hardt. Performative prediction. *arXiv preprint arXiv:2002.06673*, 2020. URL <https://arxiv.org/abs/2002.06673>.
- Georgios Piliouras and Fang-Yi Yu. Multi-agent performative prediction: From global stability and optimality to chaos. *arXiv preprint arXiv:2201.10483*, 2022. URL <https://arxiv.org/abs/2201.10483>.
- Julian Quevedo, Ansh Kumar Sharma, Yixiang Sun, Varad Suryavanshi, Percy Liang, and Sherry Yang. Worldgym: World model as an environment for policy evaluation, 2025. URL <https://arxiv.org/abs/2506.00613>.
- Burr Settles. Active learning literature survey. Technical Report 1375, University of Wisconsin–Madison, 2009. URL <https://minds.wisconsin.edu/handle/1793/60660>.
- Zhilin Wang, Han Song, Runzhe Zhan, Jusen Du, Jiacheng Chen, Tianle Li, Qingyu Yin, Yulun Wu, Zhennan Shen, Tong Zhu, Yanshu Li, Guanjie Chen, Derek F. Wong, Yafu Li, Yu Cheng, and Yang Yang. Evopolicygym: Evaluating autonomous policy evolution in interactive environments, 2026. URL <https://arxiv.org/abs/2607.02440>.
- Tianxin Wei, Zhan Shi, Minhua Lin, Bing He, Zewen Liu, Yisi Sang, Yuanchen Bei, Xuying Ning, Jiaru Zou, Ting-Wei Li, Xiao Lin, Yanjun Zhao, Chi Wang, Benoit Dumoulin, Dakuo Wang, Jingrui He, and Hanqing Lu. Evo-harness: Context-to-harness skill compilation for self-evolving agents. *arXiv preprint arXiv:2608.15071*, 2026. URL <https://arxiv.org/abs/2608.15071>.
- Ting Wu, Xuefeng Li, and Pengfei Liu. Progress or regress? self-improvement reversal in post-training. *arXiv preprint arXiv:2407.05013*, 2024. URL <https://arxiv.org/abs/2407.05013>. ICLR 2025 Conference Submission 8799 (OpenReview forum RFqeoVfLHa).
- Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin godel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025. URL <https://arxiv.org/abs/2505.22954>.
- Yanzhi Zhang, Yu Ma, Yilin Cheng, Jian Li, and Yitong Duan. M3: A state-event generative foundation model for market microstructure dynamics. *arXiv preprint*, 2026. URL [https://arthurzhang02.github.io/m3-market-microstructure/M3\\_paper.pdf](https://arthurzhang02.github.io/m3-market-microstructure/M3_paper.pdf).

## A EXTENDED FIGURES AND ARTIFACT NOTES

The release includes the frozen source snapshot, confirmatory result manifests, and generated figure bundles. Every figure bundle contains PDF, SVG, PNG, CSV, Parquet, metadata, source hashes, and the analysis script. The metadata records the unit of inference and distinguishes descriptive spans from bootstrap intervals.

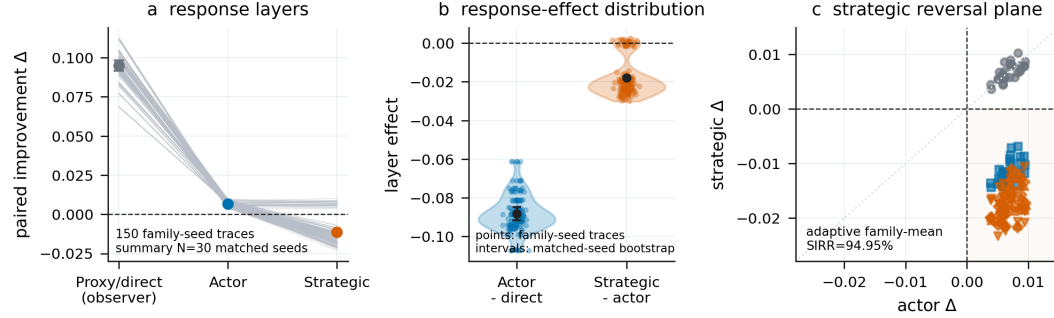


Figure 6: World-response decomposition. Panel a traces 150 family-by-seed observations (five opponent families  $\times$  30 matched seeds) through proxy/direct observer, actor, and strategic worlds; its summary intervals use the 30 matched seeds. Panel b separates mechanical and strategic effects; Panel c marks the strategic reversal quadrant.

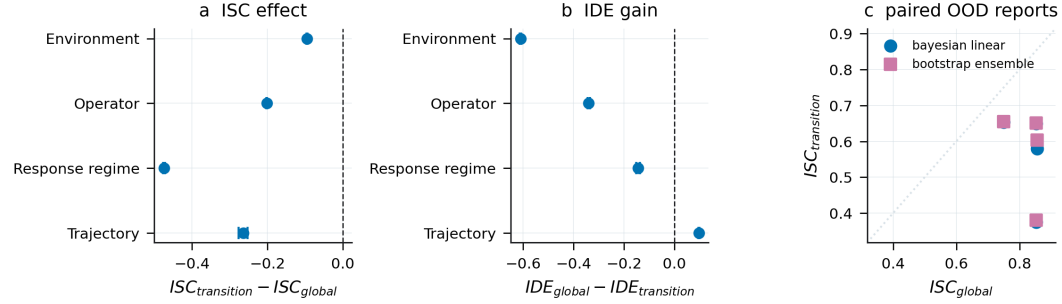


Figure 7: Powered OOD negative result. Forest horizontal ranges are descriptive spans across the two fitted model families in each split, not confidence intervals; the scatter retains the paired reports and the diagonal indicates equal ISC.

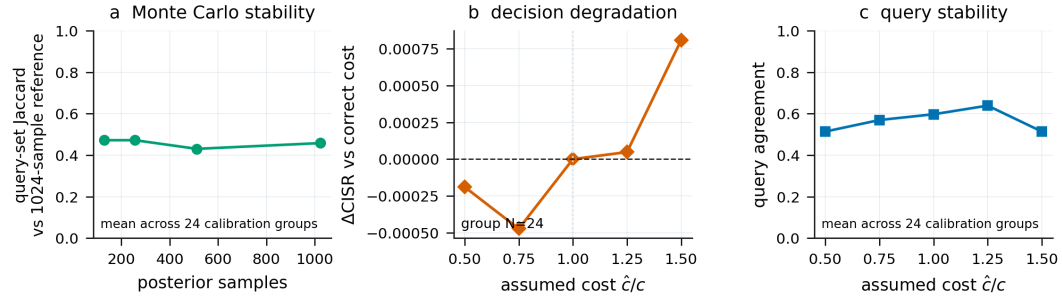


Figure 8: PIVOT-KG robustness to posterior sample count and cost misspecification. The curves report means over the registered calibration groups; they are numerical-stability diagnostics, not additional superiority claims.

**Metric and claim boundary.** The canonical metric implementation is the frozen statistics module. The closed-loop CTI/CISR values sum over rounds; the budget frontier is a one-set ( $T = 1$ ) selection problem. Their scales are therefore reported separately. The external adaptive and finance artifacts remain bounded evidence; no claim is promoted merely because a plot is visually strong. Internal experiment identifiers, configuration digests, and exact source hashes are retained in the supplement

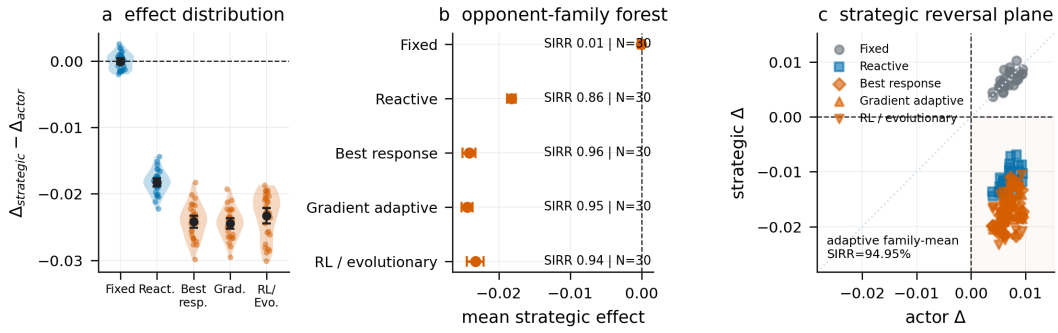


Figure 9: Strategic generalization across opponent mechanisms. Raincloud and forest views show family-specific effects and SIRR, each over 30 matched opponent-seed clusters. The reversal plane labels the unweighted mean over the three adaptive family-level SIRRs; it uses both color and marker shape so the failure is legible in grayscale.

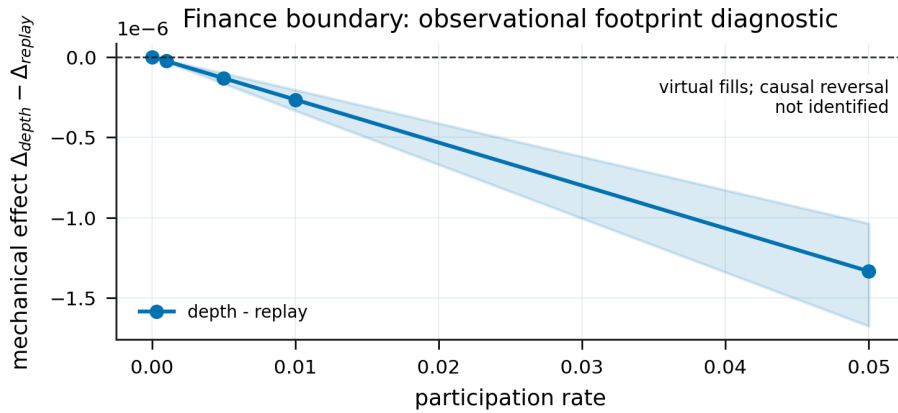


Figure 10: Initial single-asset public finance boundary diagnostic over participation. Shading is a seven-session descriptive interval; fills are virtual and the figure does not identify causal impact. The separate 12-session, three-asset expansion is summarized in the public-data table.

and figure metadata for audit. The raw sampled reversal-rate cells used in the operator-shift display remain available alongside the complete compressed transition stream.

Table 2: Controlled and public evidence summary.

| Evidence layer         | Estimate | 95% interval       | Unit / interpretation                   |
|------------------------|----------|--------------------|-----------------------------------------|
| Operator shift         | 0.2627   | [0.1684, 0.3763]   | IDE contrast; paired transition         |
| Closed-loop validation | 1.4883   | [1.0618, 1.9105]   | CISR reduction; trajectory seed         |
| OOD evaluator contrast | -0.2590  | [-0.3588, -0.1707] | transition ISC minus global ISC; null   |
| Strategic adaptation   | -0.0240  | [-0.0249, -0.0231] | strategic minus actor; opponent cluster |

Table 3: Public-data boundary summary.

| Public-data boundary                  | Value     |
|---------------------------------------|-----------|
| Initial single-asset sessions         | 7         |
| Expanded primary sessions             | 12        |
| Expanded primary reversals / eligible | 0/7       |
| Expanded holdout reversals / eligible | 0/5       |
| Expanded pooled depth effect          | -4.16e-07 |
| Causal impact identified              | No        |